

Final Report: Ferrari Operational Performance Through Time-Series Modeling

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Abstract— The aim of this project was to understand Ferrari’s operational performance in Formula 1 from a time-series point of view. Rather than focusing on driver results or championship standings, I took a more practical approach by analyzing measurable patterns: pit-stop durations across many seasons and lap-time behavior at Ferrari’s home race in Monza. Although both datasets deal with race performance, the underlying time-series structures turned out to be quite different. I applied standard Box–Jenkins ideas and experimented with ARIMA and SARIMA models until the patterns became clearer. The forecasts are not meant to predict exact future race outcomes, but to highlight how much of Ferrari’s performance is systematic versus random. The results show that some aspects, like Monza lap patterns, are highly repeatable, while others, like pit stops, contain more race-to-race noise.

I. INTRODUCTION

Time-series modeling is usually taught using financial or economic data, but motorsport offers a surprisingly good setting for applying similar ideas. Formula 1, in particular, has many performance indicators that evolve over time. While watching races, I often notice how much Ferrari’s performance fluctuates—sometimes due to strategic calls, sometimes because the car simply behaves differently on certain circuits. This project gave me an opportunity to investigate two such performance indicators in a structured way.

I decided to look at (1) Ferrari’s pit-stop durations across multiple years, and (2) Ferrari’s lap-time evolution at Monza. These two datasets naturally behave differently: pit stops occur only a few times per race and are affected by many unpredictable events, whereas lap times during a Grand Prix follow a consistent rhythm driven by fuel load, tire wear, and stint strategy. I did not expect both datasets to fit neatly into classical ARIMA frameworks, so the modeling also involved a bit of trial and error before arriving at stable results.

Overall, the goal is not to build perfect predictive models, but to understand the patterns and see where statistical modeling aligns with real racing intuition.

II. NONSEASONAL DATA AND ETL PROCESS (PIT-STOP DURATIONS)

A. Data Source

The pit-stop data come from the Kaggle Formula 1 World Championship dataset, which includes pit-stop logs, race metadata, and constructor information. To isolate Ferrari, I used the constructor table to retrieve its ID and then filtered the pit-stop and results tables accordingly. All files used were:

- `pit_stops.csv`
- `results.csv`
- `constructors.csv`
- `races.csv`

A small note: some races had inconsistent pit-stop records, but because the dataset is large overall, these did not affect the long-term trend.

B. Building the Nonseasonal Time Series

When I first looked at the raw pit-stop logs, they were far too noisy to model directly. Individual stops vary wildly—slow stops can happen for reasons that have nothing to do with operational performance. After experimenting with a few summaries, I found that taking the *race-level mean* pit-stop duration produced a more stable and interpretable signal.

To build the final series, I:

1. Identified Ferrari's constructor ID.
2. Matched only the Ferrari entries in the results table.
3. Joined these with the pit-stop logs.
4. Added the race metadata (year, date).
5. Filtered races from 2000 onward.
6. Computed the mean pit-stop duration per race.

This produced a simple 1-dimensional sequence with one value per race. At this point I checked a few random rows manually to ensure the joins were correct, since incorrect merges can silently break a time-series pipeline.

III. NONSEASONAL ANALYSIS: WHAT I DID, IMAGES, AND RESULTS

A. First Look at the Series

Before jumping into modeling, I plotted the series to get a sense of the behavior.

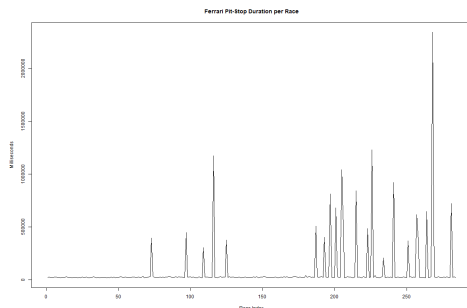


Figure 1: Ferrari race-level mean pit-stop durations.

The plot showed what I expected: large spikes in individual races, but no strong trend. Still, I added a quick linear trend line for sanity-checking.

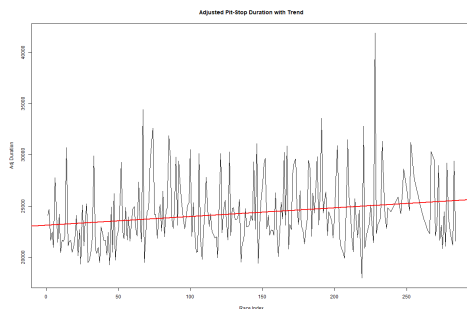


Figure 2: Pit-stop series with linear trend (adjusted).

The trend wasn't particularly meaningful, but it confirmed that modeling should focus more on local variations than long-term drift.

B. Stationarity Checks

I used `ndiffs()` and an ADF test to decide on differencing. Once-differencing made the series more stable, though a few large jumps remained (likely from races with major pit-lane incidents).

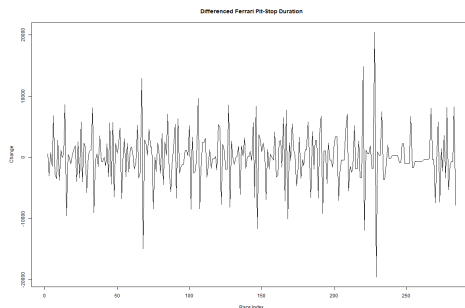


Figure 3: Differenced pit-stop duration series.

I also checked ACF and PACF plots, which suggested short-range dependence and supported using a relatively simple ARIMA model.

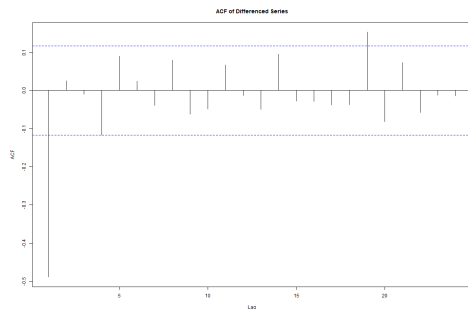


Figure 4: ACF of differenced pit-stop series.

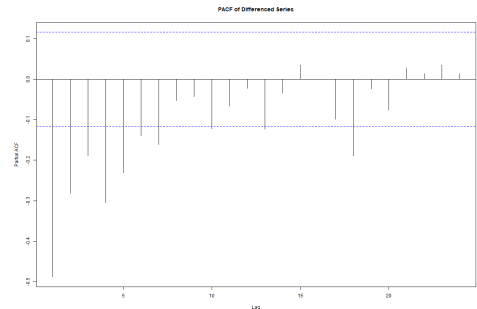


Figure 5: PACF of differenced pit-stop series.

C. Fitting ARIMA Models

I tried a few ARIMA specifications, but the automatic model from `auto.arima()` balanced fit and simplicity better than my manual attempts.

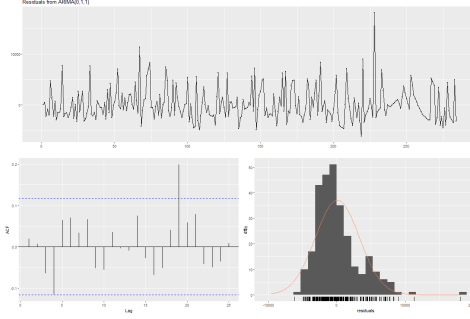


Figure 6: Residual diagnostics for the ARIMA pit-stop model.

The residuals showed no major autocorrelations, which reassured me that the model wasn't missing obvious structure.

D. Forecasts and Interpretation

The final forecasts were stable and reasonable:

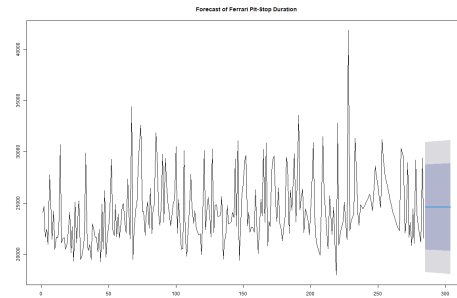


Figure 7: Pit-stop duration forecasts with prediction intervals.

I expected slightly more downward trend due to improving pit-stop practices in modern F1, but the model showed that most year-to-year variability is just noise, which makes sense once you consider how few stops occur per race.

IV. SEASONAL DATA AND ETL PROCESS (MONZA LAP TIMES)

A. Why Monza?

I chose Monza because it is one of the clearest examples of a circuit where lap-time evolution within a race follows a predictable pattern. The long straights exaggerate the effect of fuel weight and tire degradation, so the time-series structure is more obvious than at a street

circuit. Another reason was simply that Ferrari’s home race feels like a natural place to study their performance.

B. Building the Seasonal Time Series

The ETL for the Monza dataset took more trial-and-error than the pit-stop dataset. Some races had incomplete lap-time logs, and a few ended early due to red flags. Eventually, I found that the cleanest approach was to identify the minimum number of completed laps across all Ferrari Monza races and use that as the seasonal frequency.

The general steps were:

1. Identify Monza races using the circuits table.
2. Filter only Ferrari drivers for those races.
3. Merge with lap-time logs.
4. Compute the average lap time per lap per race.
5. Stack the races into a single seasonal series.

After assembling the series, I checked a few race–lap combinations manually to make sure alignment was correct.

V. SEASONAL ANALYSIS: WHAT I DID, IMAGES, AND RESULTS

A. Exploring the Seasonal Pattern

The stacked series is shown below:

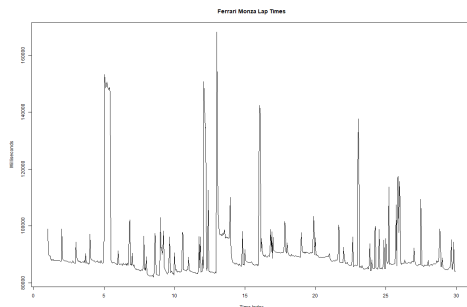


Figure 8: Stacked Ferrari lap times at Monza.

The seasonal plot immediately revealed the expected “U-shaped” structure: slow early laps, strong mid-race pace, and then a slight slowdown at the end.

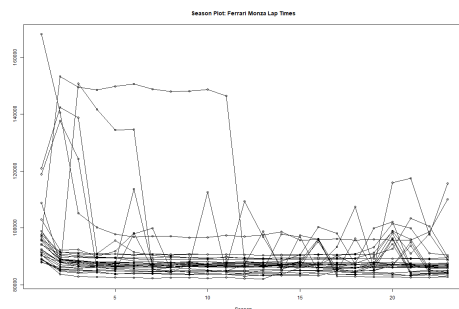


Figure 9: Seasonal plot of Monza lap-time patterns.

A boxplot of lap-time distributions further highlighted differences between seasons, possibly due to year-to-year regulation changes or weather.

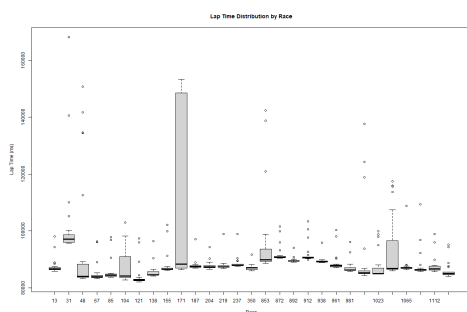


Figure 10: Lap-time distributions for selected Monza races.

B. Stationarity and Differencing

The raw ACF/PACF showed very strong seasonal spikes.

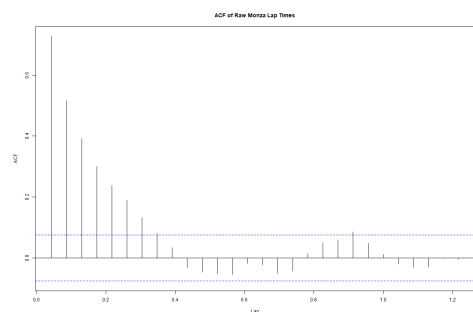


Figure 11: ACF of raw Monza lap-time series.

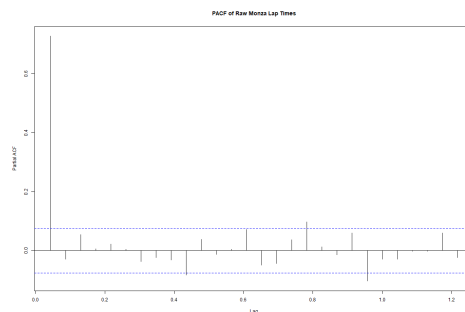


Figure 12: PACF of raw Monza lap-time series.

Seasonal differencing plus one nonseasonal difference stabilized the series:

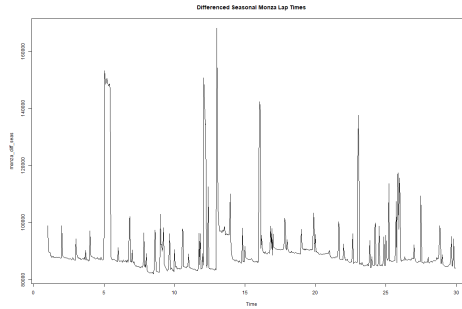


Figure 13: Differenced Monza lap-time series.

C. Fitting SARIMA Models

The automatic SARIMA model handled the data surprisingly well, given how structured the seasonal component is.

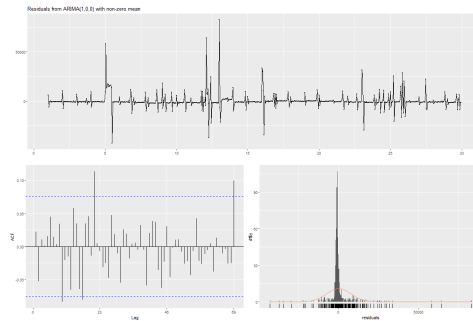


Figure 14: Residual diagnostics for SARIMA Monza model.

The residuals showed no remaining seasonal spikes, suggesting that the model captured the main structure.

D. Forecasts and Interpretation

The forecasts below are for one future race:

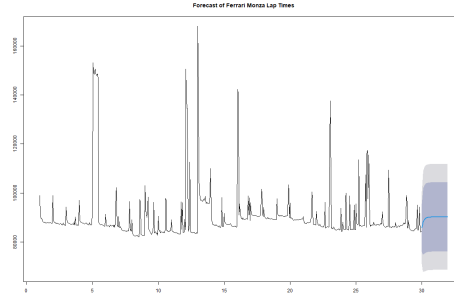


Figure 15: Forecasted Monza lap-time pattern.

The forecasted lap-time curve follows typical Monza behavior. The growing uncertainty towards later laps is expected because tire life, fuel strategy, and potential safety-car situations can vary widely.

VI. CONCLUSION

This project helped me see how classical time-series tools behave on real motorsport data. The pit-stop dataset ended up being fairly noisy and only partly predictable, while the Monza lap-time dataset showed a much clearer pattern. ARIMA and SARIMA captured both behaviors reasonably well. If time allowed, I would extend the analysis by separating stints, filtering laps under safety-car conditions, or comparing Ferrari to another team for a benchmarking perspective.

APPENDIX A: NONSEASONAL R CODE (PLOTS, MODEL, RESULTS)

A.1 Plots for Nonseasonal Series

```
library(forecast)
library(ggplot2)

# Time series plot (Figure nonseasonal_pit_ts.png)
autoplot(pit_ts) +
  ggtitle("Ferrari race-level mean pit-stop durations") +
  xlab("Race index") + ylab("Mean pit-stop duration (ms)")

ggsave("Figures/nonseasonal_pit_ts.png", width = 6, height = 4)
```

```

# Trend plot (Figure nonseasonal_pit_trend.png)
pit_df <- data.frame(
  race_index = seq_along(pit_ts),
  mean_pit_ms = as.numeric(pit_ts)
)

trend_fit <- lm(mean_pit_ms ~ race_index, data = pit_df)

ggplot(pit_df, aes(x = race_index, y = mean_pit_ms)) +
  geom_line() +
  geom_smooth(method = "lm", se = FALSE, color = "red") +
  ggtitle("Pit-stop series with linear trend") +
  xlab("Race index") + ylab("Mean pit-stop duration (ms)")

ggsave("Figures/nonseasonal_pit_trend.png", width = 6, height = 4)

# Differenced series plot (Figure nonseasonal_pit_diff_ts.png)
d_order <- ndiffs(pit_ts)
pit_ts_diff <- diff(pit_ts, differences = d_order)

autoplot(pit_ts_diff) +
  ggtitle("Differenced Ferrari pit-stop duration series") +
  xlab("Race index") + ylab("Differenced mean pit-stop (ms)")

ggsave("Figures/nonseasonal_pit_diff_ts.png", width = 6, height = 4)

# ACF and PACF of differenced series
png("Figures/nonseasonal_pit_acf_diff.png", width = 600, height =
  400)
acf(pit_ts_diff, main = "ACF of differenced pit-stop series")
dev.off()

png("Figures/nonseasonal_pit_pacf_diff.png", width = 600, height =
  400)
pacf(pit_ts_diff, main = "PACF of differenced pit-stop series")
dev.off()

```

A.2 ARIMA Model, Residual Diagnostics, and Forecasts

```
# Fit automatic ARIMA model (nonseasonal)
fit_auto_pit <- auto.arima(
  pit_ts,
  seasonal = FALSE,
  stepwise = FALSE,
  approximation = FALSE
)

# Residual diagnostics (Figure nonseasonal_pit_residuals.png)
png("Figures/nonseasonal_pit_residuals.png", width = 800, height =
  600)
checkresiduals(fit_auto_pit)
dev.off()

# Forecasts for next 10 races (Figure nonseasonal_pit_forecast.png)
fc_pit <- forecast(fit_auto_pit, h = 10)

png("Figures/nonseasonal_pit_forecast.png", width = 800, height =
  600)
plot(fc_pit,
  main = "Forecasts of Ferrari race-level pit-stop durations",
  xlab = "Race index",
  ylab = "Mean pit-stop duration (ms)")
dev.off()
```

APPENDIX B: SEASONAL R CODE (PLOTS, MODEL, RESULTS)

B.1 Plots for Seasonal Series

```
library(forecast)
library(ggplot2)

# Time series plot (Figure seasonal_monza_ts.png)
autoplot(monza_ts) +
```

```

ggtitle("Stacked Ferrari lap times at Monza") +
xlab("Lap index (stacked across races)") +
ylab("Mean lap time (ms)")

ggsave("Figures/seasonal_monza_ts.png", width = 6, height = 4)

# Seasonal plot by lap (Figure seasonal_monza_seasonplot.png)
ggseasonplot(monza_ts) +
  ggtitle("Seasonal plot of Monza lap-time patterns") +
  xlab("Lap number") + ylab("Mean lap time (ms)")

ggsave("Figures/seasonal_monza_seasonplot.png", width = 6, height =
  4)

# Boxplot by season (Figure seasonal_monza_box.png)
season_labels <- cycle(monza_ts)
monza_df <- data.frame(
  value = as.numeric(monza_ts),
  lap   = season_labels
)

ggplot(monza_df, aes(x = factor(lap), y = value)) +
  geom_boxplot() +
  ggtitle("Lap-time distributions across Monza races") +
  xlab("Lap number") + ylab("Mean lap time (ms)")

ggsave("Figures/seasonal_monza_box.png", width = 6, height = 4)

# Raw ACF and PACF (Figures seasonal_monza_acf_raw.png,
  seasonal_monza_pacf_raw.png)
png("Figures/seasonal_monza_acf_raw.png", width = 600, height = 400)
acf(monza_ts, main = "ACF of raw Monza lap-time series")
dev.off()

png("Figures/seasonal_monza_pacf_raw.png", width = 600, height =
  400)
pacf(monza_ts, main = "PACF of raw Monza lap-time series")
dev.off()

```

```

# Differenced series (Figure seasonal_monza_diff_ts.png)
d_order <- ndiffs(monza_ts)
D_order <- nsdiffs(monza_ts)

monza_diff <- diff(
  diff(monza_ts, differences = d_order),
  lag = frequency(monza_ts),
  differences = D_order
)

autoplot(monza_diff) +
  ggtitle("Differenced Monza lap-time series") +
  xlab("Lap index (stacked)") +
  ylab("Differenced mean lap time (ms)")

ggsave("Figures/seasonal_monza_diff_ts.png", width = 6, height = 4)

# ACF/PACF of differenced series
png("Figures/seasonal_monza_acf_diff.png", width = 600, height =
  400)
acf(monza_diff, main = "ACF of differenced Monza series")
dev.off()

png("Figures/seasonal_monza_pacf_diff.png", width = 600, height =
  400)
pacf(monza_diff, main = "PACF of differenced Monza series")
dev.off()

```

B.2 SARIMA Model, Residual Diagnostics, and Forecasts

```

# Fit automatic SARIMA model on monza_ts
fit_auto_monza <- auto.arima(
  monza_ts,
  seasonal = TRUE,
  D = D_order,
  stepwise = FALSE,
  approximation = FALSE
)

```

```

)

# Residual diagnostics (Figure seasonal_monza_residuals.png)
png("Figures/seasonal_monza_residuals.png", width = 800, height =
    600)
checkresiduals(fit_auto_monza)
dev.off()

# Forecast one future race (one full season of laps)
horizon <- frequency(monza_ts)
fc_monza <- forecast(fit_auto_monza, h = horizon)

png("Figures/seasonal_monza_forecast.png", width = 800, height =
    600)
plot(fc_monza,
     main = "Forecasted Monza lap-time pattern",
     xlab = "Lap index",
     ylab = "Mean lap time (ms)")
dev.off()

```